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Hidden Alpha

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1 Executive overview

The paper studies whether **mutual fund managers earn abnormal returns by trading on private information obtained through concealed personal relationships**. It constructs a novel, large-scale **hidden social network** using information scraped from more than 100,000 Facebook profiles (and their tens of millions of friends), identifying personal ties between **fund managers** and **public-firm officers**. The key empirical insight is that these ties can be *publicly visible* or *concealed* (hidden) on Facebook, creating a cross-sectional dimension—**friendship visibility**—that the paper uses to test whether concealment is related to informational value. Using holdings-based portfolio replications, trade-based timing tests, event-based return decompositions, and cross-sectional predictive regressions, the paper finds large and robust **alphas concentrated in the most hidden connections**, consistent with an information flow channel rather than mechanical style exposure or endogenous matching.

2 Main contributions

- (1) **Data innovation:** builds a new dataset of fund-manager–firm-officer social ties from Facebook and introduces the notion of *concealed* ties.
- (2) **Core fact:** connected stocks outperform *only* when the friendship is hidden; the most hidden ties generate about **135 bps/month four-factor alpha** ($t \approx 3.5$), i.e. **16%+ annually** (Table 6).
- (3) **Mechanism evidence:** managers display **timing ability**—they hold connected stocks when they will outperform and avoid connected stocks that will not (Table 6, plus the illustrative example in Fig. 1).
- (4) **Event concentration:** abnormal performance is concentrated around **earnings** and **M&A** news, and stronger when the connected officer has relevant expertise (Table 6).
- (5) **External validity and falsification:** no premium in **index funds** (where selection/timing is infeasible), and the premium exists both pre- and post-Facebook’s inception (Tables 6–6).

3 Data construction and definitions

3.1 Fund universe (Morningstar Direct)

The fund sample is the universe of U.S.-domiciled **actively managed U.S. equity mutual funds** in Morningstar Direct (1984–2020), including defunct funds to mitigate survivorship bias. Index funds and non-U.S./non-equity categories are excluded.

3.2 Firm officer universe (BoardEx)

Firm officers are taken from BoardEx. For each public firm and month, the paper identifies the set of officers active at that time and then matches officers to Facebook profiles.

3.3 Facebook identification and observable fields (Tables 1–2)

For identified users (4,198 fund managers; 101,866 firm officers), the paper observes profile-level fields (Table 1) and item-level activity objects (Table 2), including:

- friends lists (when public), backlink-based friend inference, and reaction-based inference;
- photos and engagements (likes, comments, tags) that help infer concealed ties;
- demographics and education/work history (when visible).

3.4 Friendship visibility: the key state variable

For a fund manager m and firm officer o , define two indicators:

$$V_{m,o}^{(m)} = \mathbf{1}\{\text{friendship visible from manager's friends list}\},$$

$$V_{m,o}^{(o)} = \mathbf{1}\{\text{friendship visible from officer's friends list}\}.$$

Friendship visibility is a four-way partition:

$$\text{FullyVisible: } (V^{(m)} = 1, V^{(o)} = 1), \quad \text{FirmOfcrHides: } (V^{(m)} = 1, V^{(o)} = 0),$$

$$\text{FundMgrHides: } (V^{(m)} = 0, V^{(o)} = 1), \quad \text{FullyHidden: } (V^{(m)} = 0, V^{(o)} = 0).$$

Economically, the paper interprets these categories as increasing degrees of **concealment** of the social tie.

3.5 Tradable vs traded friendships (Table 4)

A manager–officer friendship is:

- **tradable** if manager tenure overlaps officer tenure and the firm’s stock is held by at least one fund in the same Morningstar category in that month;
- **traded** if the manager’s fund actually holds the firm’s stock during officer tenure.

Table 4 shows how the number of pairs, managers, officers, and traded observations varies across visibility buckets.

3.6 Interaction intensity (Table 9)

Define

$$Interact_{m,o,t-1} = \mathbf{1}\{\text{manager and officer interacted on Facebook (likes/comments/tags) in the prior year}\}.$$

The paper uses this to test whether more “active” ties are more informative.

4 All models and empirical objects

4.1 Model 1: Portfolio tilting into connected stocks (Table 5; Eq. (1) in the paper)

The unit of observation is stock–fund–month. Let $w_{f,k,t}$ be the portfolio weight:

$$w_{f,k,t} = \frac{\text{dollar holding of stock } k \text{ by fund } f \text{ at } t}{\text{fund TNA at } t}.$$

For each stock–fund–month, define the visibility dummies $\{FV, FOH, FMH, FH\}$ corresponding to *FullyVisible*, *FirmOfcrHides*, *FundMgrHides*, *FullyHidden*. The baseline tilt regression is:

$$w_{f,k,t} = \alpha + \beta_{FV} FV_{f,k,t} + \beta_{FOH} FOH_{f,k,t} + \beta_{FMH} FMH_{f,k,t} + \beta_{FH} FH_{f,k,t} + \Gamma' X_{f,k,t} + \mu_t + \mu_f + \mu_k + \varepsilon_{f,k,t}. \quad (1)$$

Controls. $X_{f,k,t}$ includes (i) the Daniel–Grinblatt–Titman–Wermers *style* share and (ii) stock characteristic percentiles: pME (size), pBM (book-to-market), and $R12$ (past 12-month return), as described under Table 5.

4.2 Model 2: Calendar-time connected vs non-connected portfolio returns (Table 6)

For each fund–month, split holdings into connected and non-connected sets. Let $\Omega_{f,t}^C$ be connected holdings and $\Omega_{f,t}^{NC}$ non-connected holdings. Construct within-fund portfolio returns by weighting by dollar holdings:

$$R_{f,t}^C = \sum_{k \in \Omega_{f,t}^C} \tilde{w}_{f,k,t} r_{k,t+1}, \quad R_{f,t}^{NC} = \sum_{k \in \Omega_{f,t}^{NC}} \tilde{w}_{f,k,t} r_{k,t+1},$$

where $\tilde{w}$ is renormalized to sum to one within the subportfolio. Aggregate across funds by value-weighting with subportfolio TNA and compute:

$$R_t^{LS} = R_t^C - R_t^{NC}.$$

4.3 Model 3: Four-factor alpha estimation (monthly and daily)

For a generic portfolio return series R_t , four-factor alpha is estimated via:

$$R_t - r_t^f = \alpha + \beta_M(MKT_t) + \beta_S(SMB_t) + \beta_H(HML_t) + \beta_U(UMD_t) + u_t. \quad (2)$$

Monthly versions use monthly factors; daily versions use daily factors (as in the paper’s news-event tests).

4.4 Model 4: DGTW adjustment (Tables 6, 8, 9)

For stock k in month t ,

$$r_{k,t}^{DGTW} = r_{k,t} - r_{b(k,t),t}^{bench}$$

where $b(k, t)$ is the Daniel et al. (1997) characteristic-matched benchmark cell (size, book-to-market, and momentum). Portfolio-level DGTW-adjusted returns are formed by aggregating $r_{k,t}^{DGTW}$ using the same weights as in the raw return portfolios.

4.5 Model 5: Connected Not-Held (CNH) timing test (Table 8)

For each fund-month, define:

- **Connected Held (CH):** connected stocks actually held by fund f .
- **Connected Not-Held (CNH):** connected stocks *not* held by f but held by at least one other fund in the same Morningstar category.

Compute R_t^{CH} , R_t^{CNH} and the spread R_t^{CH-CN} . **Logic:** if results were purely due to endogenous matching (“good managers matched with good officers’ firms”), then CH and CNH should perform similarly; a CH>CNH spread implies **timing/selection conditional on the connection**.

4.6 Model 6: Trade-based buy–sell timing portfolios (Table 7)

Between two consecutive holdings reports, classify each stock as:

$$Buy_{f,k,t} = \mathbf{1}\{\text{position increases or is newly initiated}\}, \quad Sell_{f,k,t} = \mathbf{1}\{\text{position decreases or is fully exited}\}.$$

Form buy and sell portfolios (connected and non-connected), value-weight returns by *new dollars* allocated, and compute:

$$R_t^{LS} = R_t^{Buy} - R_t^{Sell},$$

with extensive-vs-intensive splits: extensive (new/fully divested) vs intensive (increased/decreased).

4.7 Model 7: News-event daily alphas (Tables 10–11)

Assign to each stock held in a fund portfolio its *next-month daily returns*. For each fund-day, split stocks into *News* vs *No News* depending on whether the firm has a news announcement that day. Compute daily four-factor alphas for CH, NCH, and CNH, and for long–short spreads. Table 11 further splits by:

- event type (earnings/revenues; M&A; etc.),
- officer expertise (CEO vs CFO vs other C-suite),
- analyst coverage (high vs low).

4.8 Model 8: Cross-sectional predictability (Fama–MacBeth; Table 12)

Define the **weight-difference signal**:

$$DiffWeight_{k,t} = \bar{w}_{k,t}^{FB} - \bar{w}_{k,t}^{nonFB},$$

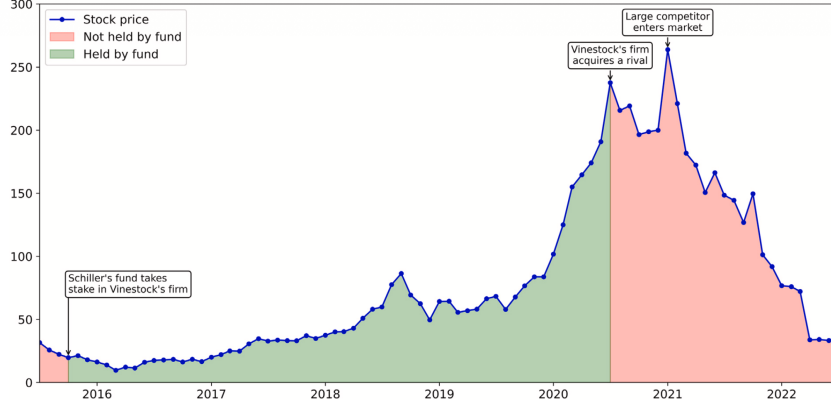


Fig. 1. Holding status of Vinestock's firm's stock by Schiller's fund.

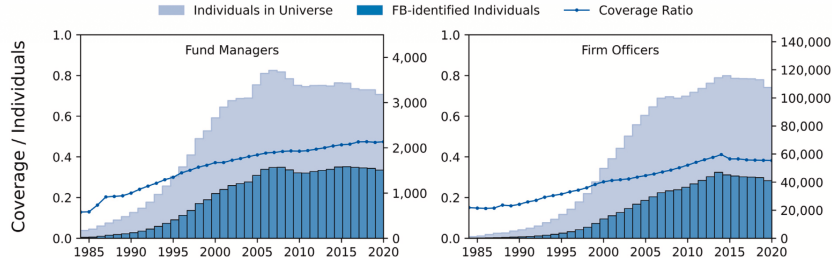


Fig. 2. Ratios of Facebook-identified fund managers and firm officers.

the difference between the average weight Facebook-connected funds allocate to stock k and the average weight all other funds allocate. Estimate a stock-specific loading from a rolling regression:

$$r_{k,t} = a_k + \beta_k(DiffWeight) DiffWeight_{k,t-1} + \varepsilon_{k,t}.$$

Then run monthly cross-sectional regressions of next-month excess returns:

$$\begin{aligned} ExcessRet_{k,t+1} = & \lambda_{0,t} + \lambda_{1,t} \beta_k(DiffWeight) + \lambda_{2,t} ME_{k,t} + \lambda_{3,t} BM_{k,t} \\ & + \lambda_{4,t} MOM_{k,t} + \lambda_{5,t} STR_{k,t} + \lambda_{6,t} IMOM_{k,t} + \lambda_{7,t} SUE_{k,t} + u_{k,t+1}. \end{aligned} \quad (3)$$

Table 12 reports the time-series averages of λ 's (Fama–MacBeth risk premia).

5 Interpretation of figures

5.1 Fig. 1: A motivating example of timing

Fig. 1 provides an illustration: a fund manager connected to a firm officer alternates between holding and not holding the officer's firm's stock, and the holding periods align with better subsequent performance. The figure is used to motivate why the paper focuses on **timing ability** rather than only unconditional connected-vs-non-connected averages.

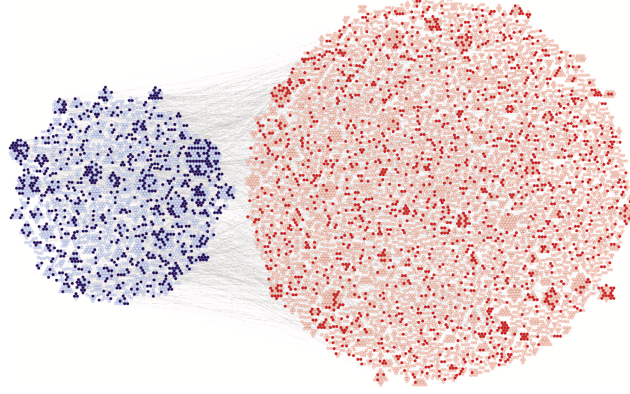


Fig. 3. Network of tradable friendships between fund managers and firm officers.

5.2 Fig. 2: Facebook coverage over time

Fig. 2 shows the annual identification ratios for fund managers and firm officers on Facebook. Coverage rises over time and is substantial in the later sample. The key takeaway is that the network is **measurable at scale** but incomplete, motivating the identification/selection checks in Table 13 and Fig. 5.

5.3 Fig. 3: The tradable friendship network

Fig. 3 visualizes the network of tradable friendships between managers and officers. The figure conveys that the network has a nontrivial structure (clusters and hubs), which makes it plausible that information can diffuse through hidden ties.

5.4 Fig. 4: Post-purchase CARs by visibility

Fig. 4 plots cumulative abnormal returns after initial purchases of connected stocks. The monotonic ordering is the key empirical signature:

$$CAR(\text{FullyHidden}) > CAR(\text{partially hidden}) > CAR(\text{FullyVisible}),$$

and the accumulation is gradual, consistent with managers *earning* information-based returns rather than capturing a one-day announcement jump.

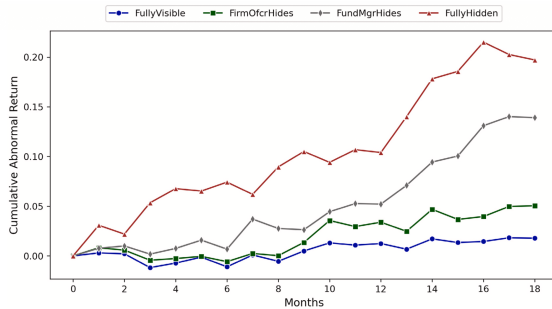


Fig. 4. CARs on connected stocks by friendship visibility.

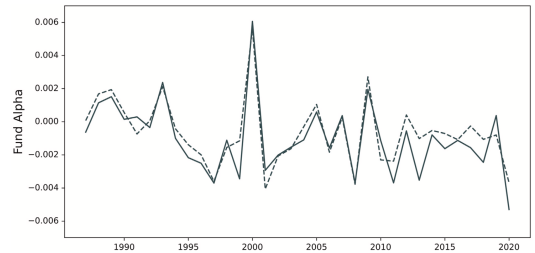


Fig. 5. Performance of Facebook-identified vs. non-identified funds.

5.5 Fig. 5: Performance of Facebook-identified vs non-identified funds

Fig. 5. This figure plots the annual value-weighted four-factor alpha of *Facebook-identified* mutual funds (dashed line) versus all other (non-identified) funds in the benchmark universe defined in Section 1.4 (solid line) over 1984–2020. The key takeaway is that *Facebook-identified* status does **not** mechanically imply superior performance, supporting the interpretation that the paper’s results are not driven by a selection of unusually skilled managers into the Facebook-identified subsample.

6 Interpretation of tables

6.1 Table 1: What Facebook fields are observable?

Table 1 summarizes which profile-level fields are observable and for what fraction of profiles. Most users have profile pictures and cover photos; friends lists are visible for about half the sample, which motivates the need for inference via backlinks and reactions.

Table 1

Profile-level Facebook data.

	Fund Managers (N Profiles = 4,198)		Firm Officers (N Profiles = 101,866)	
	% Share	N Profiles	% Share	N Profiles
Screen Name	1	4,198	1	101,866
Username	1	4,198	1	101,866
User ID	1	4,198	1	101,866
Gender	.79	3,320	.71	72,127
Date of Birth	.01	37	.02	1,884
Friends List	.56	2,353	.53	54,423
Relationship Status	.21	885	.21	21,542
Family Members	.24	988	.23	23,554
Profile Picture	.90	3,793	.94	96,153
Cover Photo	.88	3,707	.89	90,864
Other Photos	.67	2,815	.77	78,023
Work	.33	1,379	.36	36,774
College	.46	1,939	.52	52,776
High School	.37	1,548	.44	45,070
Current City	.53	2,245	.48	49,172
Hometown	.43	1,815	.43	43,428
Other Places Lived	.22	919	.17	17,818
Life Events	.25	1,054	.33	33,209

Table 2

Item-level Facebook data.

	Fund managers (N Profiles = 4,198)					Firm officers (N Profiles = 101,866)				
	Mean	Median	SD	N Items	N Profiles	Mean	Median	SD	N Items	N Profiles
Friends	269	167	394	1,095,018	4,066	341	188	635	34,148,303	100,193
Friends-Published	310	213	425	728,897	2,353	443	275	561	24,102,607	54,423
Friends-Backlinks	110	66	144	180,951	1,648	10	5	19	402,652	41,613
Friends-Reactions	126	69	199	180,058	1,425	196	111	664	7,932,927	40,540
Photos	56	6	321	187,328	3,374	110	12	420	9,831,174	89,294
Reactions	663	118	2,601	2,138,914	3,226	695	215	2,039	59,448,651	85,512
Likes	325	91	2,094	1,666,017	3,173	598	184	1,752	50,593,776	84,627
Comments	148	33	537	435,814	2,954	105	35	322	8,283,007	78,817
Tags	37	5	144	37,083	1,014	17	5	67	571,868	33,803

6.2 Table 2: Scale of item-level data (friends, photos, reactions)

Table 2 shows the data are rich at the item level: many users have hundreds of friends and substantial engagement objects. Crucially, friend links can be observed not only through public friends lists but also through backlinks and photo-reaction-based inference, which is essential for measuring *hidden* ties.

6.3 Table 3: Coverage of the Facebook-identified fund universe

Table 3 documents that the Facebook-identified fund sample covers roughly half of the Morningstar universe by fund count and TNA in a typical year, and covers a large share of the investable equity universe by market cap (close to full coverage), which supports external validity of the holdings-based analyses.

6.4 Table 4: How many friendships are hidden, tradable, and traded?

Table 4 reports the number of manager–officer pairs across the four visibility buckets. Two interpretive points:

Table 3
Summary statistics for Facebook-identified funds.

	Mean	Median	Min.	Max.	SD
Facebook-identified Funds (FB Funds)	1,126	1,508	23	1,985	755
% of Funds in MS Universe	.53	.59	.15	.75	.19
% of TNA in MS Universe	.50	.53	.09	.79	.22
Facebook-identified Fund Managers	934	1,166	22	1,584	614
% of Fund Managers in MS Universe	.35	.38	.13	.47	.11
Stocks held by FB funds	3,768	4,267	310	5,314	1,443
% of Stocks in CRSP Universe	.50	.57	.05	.70	.19
% of Market Cap. in CRSP Universe	.85	.96	.32	1	.18
Firm Officers held by FB Funds	59,949	64,050	1,122	115,811	45,273
% of Firm Officers in BoardEx Universe	.95	.99	.56	1	.10
Facebook-identified Firm Officers	22,116	18,405	169	55,596	19,903
% of Firm Officers held by FB Funds	.29	.29	.15	.52	.11

Table 4
Statistics on Facebook friendships by visibility.

	N Pairs	N Fund Mgrs.	N Firm Offcs.	N Stocks	N Obs.
<u>Friendships-All</u>	18,478	2,803	10,748	–	–
FullyVisible	6,748	1,583	4,590	–	–
FirmOfcrHides	6,211	1,438	3,968	–	–
FundMgrHides	4,235	927	3,008	–	–
FullyHidden	1,284	426	993	–	–
<u>Friendships-Tradable</u>	9,012	2,107	6,050	3,371	–
FullyVisible	3,298	1,105	2,516	2,043	–
FirmOfcrHides	3,084	1,007	2,253	1,730	–
FundMgrHides	2,038	658	1,600	1,354	–
FullyHidden	592	273	486	461	–
<u>Friendships-Traded</u>	2,908	1,052	2,143	1,510	171,755
FullyVisible	930	459	768	755	45,542
FirmOfcrHides	959	441	779	680	45,020
FundMgrHides	758	343	618	576	51,350
FullyHidden	261	136	214	245	29,495

- (i) Hidden ties are not rare: thousands of friendships are partially hidden, and hundreds are fully hidden even after imposing “tradable” restrictions.
- (ii) A sizable subset is actually traded (“Friendships–Traded”), providing enough variation for return tests and timing tests.

6.5 Table 5: Managers overweight connected stocks, more so when ties are hidden

Table 5 estimates Eq. (1). Key facts (coefficients in basis points):

- Baseline average weight is about 74.6 bps.
- Incremental overweight increases with concealment: ≈ 35 bps (FullyVisible), 64 bps (FirmOfcrHides), 96 bps (FundMgrHides), and 136 bps (FullyHidden) in the simplest specifications.
- With fund and firm fixed effects and characteristic/style controls, FullyHidden remains economically and statistically significant.

Interpretation: hidden ties are **behaviorally relevant** (managers tilt capital toward them), setting up the performance tests.

Table 5
Portfolio weights in connected stocks by visibility.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	74.59*** [3.21]	74.59*** [3.21]	74.59*** [3.21]	74.59*** [3.21]	1.42 [2.82]	1.43 [2.83]	–24.80*** [4.05]	–24.83*** [4.06]
FullyVisible	32.18*** [7.06]				1.05 [1.71]		8.68* [5.10]	
FirmOfcrHides		63.88*** [8.67]						
FundMgrHides			95.80*** [13.22]					
FullyHidden				136.47*** [15.34]		50.94*** [7.57]		21.64*** [5.46]
Controls	No	No	No	No	Yes	Yes	Yes	Yes
Fixed Effects	Period	Period	Period	Period	Period	Period	Period	Period
Adj. R Squared	0.01	0.01	0.01	0.01	0.36	0.36	0.40	0.40
N ($\times 10^6$)	34.49	34.49	34.48	34.47	31.28	31.27	31.28	31.27

Table 6
Returns on connected stocks by visibility.

	Raw Return			Four-Factor Alpha			DGTW-Adjusted		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	1.05*** (3.07)	0.98*** (3.92)	0.07 (0.28)	0.11 (0.43)	–0.00 (–0.14)	0.11 (0.44)	0.15 (0.70)	0.03 (0.63)	0.12 (0.57)
FirmOfcrHides	1.18*** (3.72)	0.98*** (0.74)	0.20 (0.74)	0.30 (1.12)	–0.01 (–0.43)	0.32 (1.16)	0.34 (1.62)	0.02 (0.41)	0.32 (1.53)
FundMgrHides	1.53*** (3.73)	0.95*** (3.99)	0.57* (1.87)	0.56* (1.81)	–0.00 (–0.02)	0.56* (1.83)	0.71*** (2.80)	0.03 (0.67)	0.69*** (2.74)
FullyHidden	2.40*** (5.02)	0.93*** (3.62)	1.48*** (3.88)	1.35*** (3.54)	–0.00 (–0.16)	1.36*** (3.57)	1.39*** (3.84)	0.03 (0.57)	1.37*** (3.80)

6.6 Table 6: Main result—large alpha concentrated in FullyHidden ties

Table 6 forms calendar-time portfolios and estimates returns and alphas. The core monotonic pattern:

- Connected vs non-connected: non-connected returns are roughly similar across buckets (around 0.93–0.98% raw per month), but connected returns rise sharply with concealment.

Table 7
Trade-based returns on connected stocks.

	Connected			Non-Connected			
	(1) Buy	(2) Sell	(3) LS	(4) Buy	(5) Sell	(6) LS	(3)–(6) Spread
FullyVisible							
All Trades	0.32 (0.54)	−0.18 (−0.31)	0.46 (0.55)	0.05 (0.37)	0.16 (1.26)	−0.11 (−0.77)	0.57 (0.67)
Extensive Trades	0.30 (0.35)	−0.27 (−0.30)	0.60 (0.57)	0.15 (0.97)	0.17 (1.00)	−0.04 (−0.20)	0.64 (0.60)
Intensive Trades	0.35 (0.61)	−0.11 (−0.25)	0.46 (0.66)	0.08 (0.52)	0.16 (1.18)	−0.09 (−0.56)	0.55 (0.75)
FirmOfcrHides							
All Trades	0.48 (0.94)	−0.23 (−0.45)	0.67 (1.02)	0.02 (0.12)	0.11 (0.92)	−0.11 (−0.75)	0.78 (1.17)
Extensive Trades	0.40 (0.51)	−0.56 (−0.70)	0.80 (0.81)	0.18 (0.96)	0.25 (1.62)	0.05 (0.23)	0.75 (0.79)
Intensive Trades	0.51 (0.96)	−0.19 (−0.39)	0.69 (1.10)	0.02 (0.12)	0.00 (0.03)	−0.02 (−0.12)	0.71 (1.07)
FundMgrHides							
All Trades	0.77** (2.14)	−0.61* (−1.94)	1.36*** (3.15)	−0.09 (−1.09)	−0.10 (−0.91)	−0.01 (−0.04)	1.36*** (3.01)
Extensive Trades	1.45** (2.00)	−1.04* (−1.85)	2.29*** (3.06)	−0.06 (−0.43)	−0.05 (−0.38)	0.12 (0.66)	2.17*** (2.86)
Intensive Trades	0.69* (1.81)	−0.28 (−0.89)	0.95** (2.10)	−0.05 (−0.55)	−0.09 (−0.98)	0.03 (0.30)	0.92** (2.02)
FullyHidden							
All Trades	1.27*** (2.75)	−0.68 (−1.34)	1.89*** (2.83)	−0.04 (−0.34)	0.01 (0.06)	−0.04 (−0.31)	1.93*** (2.89)
Extensive Trades	2.25** (2.21)	−1.90* (−1.84)	2.76*** (2.79)	−0.11 (−0.71)	−0.21 (−1.27)	0.02 (0.14)	2.74*** (2.78)
Intensive Trades	1.12** (2.42)	−0.53 (−1.20)	1.56** (2.58)	0.02 (0.18)	0.15 (1.51)	−0.13 (−0.96)	1.68*** (2.75)

- The long–short four-factor alpha is essentially zero for FullyVisible and FirmOfcrHides, modest for FundMgrHides, and **very large for FullyHidden**: $\alpha^{LS} \approx 1.36\%$ per month with $t \approx 3.57$.

This is the paper’s headline “hidden alpha” finding.

6.7 Table 7: Trade-based timing—buy connected winners, sell connected losers

Table 7 assigns holdings changes into buy vs sell and computes buy–sell long–short alphas. Interpretation:

- For hidden buckets (FundMgrHides and FullyHidden), buy portfolios earn positive alphas and sell portfolios earn negative alphas, producing large buy–sell spreads.
- The spread is absent for non-connected trades, consistent with information being specific to hidden ties rather than generic trading skill.
- Results hold for both extensive and intensive margin trades, suggesting timing skill in both entering/exiting and scaling positions.

6.8 Table 8: Connected Held vs Connected Not-Held—a key falsification

Table 8 compares held connected stocks (CH) to connected stocks not held by the fund (CNH) but held by peers. The CH–CNH spread is **largest for FullyHidden** (about 1.19% four-factor alpha per month). Interpretation: the alpha is not simply due to “connected firms are good”; managers appear to know *when* to hold connected stocks.

Table 8
Returns on connected non-held stocks.

	Connected Held (CH)			Connected Not-Held (CNH)			Long CH/Short CNH		
	Raw	Alpha	DGTV	Raw	Alpha	DGTV	Raw	Alpha	DGTV
FullyVisible	1.05*** (3.07)	0.11 (0.43)	0.15 (0.70)	1.12*** (3.09)	0.06 (0.30)	0.05 (0.15)	0.05 (0.35)	0.11 (0.63)	0.17 (0.63)
FirmOfcHides	1.18*** (3.43)	0.30 (1.12)	0.34 (1.42)	1.18*** (3.44)	0.09 (0.36)	0.16 (0.77)	-0.09 (-0.36)	0.10 (0.31)	0.08 (0.30)
FundMgrHides	1.53*** (3.73)	0.36* (1.81)	0.71*** (2.80)	0.98*** (2.93)	-0.02 (-0.09)	0.01 (0.05)	0.54 (1.59)	0.57* (1.66)	0.71** (2.37)
FullyHidden	1.30*** (5.02)	0.30*** (3.54)	0.60*** (3.84)	1.10*** (2.41)	0.16 (0.50)	0.20 (0.67)	1.30*** (2.75)	1.19*** (2.59)	1.30*** (2.73)

Table 9
Returns on connected stocks by interaction intensity.

	Raw Returns			Four-Factor Alpha			DGTV-Adjusted		
	Interacted	Non-Inter	LS	Interacted	Non-Inter	LS	Interacted	Non-Inter	LS
FullyVisible	1.12*** (2.86)	1.07*** (3.03)	0.05 (0.25)	0.15 (0.41)	0.13 (0.39)	0.02 (0.20)	0.13 (0.34)	0.18 (-0.19)	-0.05 (-0.19)
FirmOfcHides	1.29*** (2.99)	1.18*** (3.41)	0.12 (0.33)	0.38 (1.30)	0.33 (1.19)	0.05 (0.51)	0.39 (1.16)	0.37 (1.12)	0.02 (0.00)
FundMgrHides	2.30*** (4.03)	1.47*** (3.41)	0.89* (1.87)	1.36** (3.08)	0.56* (1.70)	0.81* (1.82)	1.21*** (2.67)	0.68* (1.45)	0.53* (1.69)
FullyHidden	3.42*** (4.41)	2.32*** (4.89)	1.11*** (2.16)	2.35*** (3.80)	1.20*** (2.56)	1.04*** (2.43)	2.44*** (3.94)	1.38*** (2.68)	1.04*** (2.59)

Table 10
Returns on connected stocks around news.

	Connected Held (CH)		Non-Connected Held (NCH)		Connected Not-Held (CNH)		Long CH/Short NCH		Long CH/Short CNH	
	News	No News	News	No News	News	No News	News	No News	News	No News
FullyVisible	0.001** (1.88)	0.006 (0.76)	0.019*** (3.94)	0.004 (1.00)	0.019** (2.21)	-0.006 (-0.75)	0.012 (0.40)	0.002 (0.24)	0.012 (0.44)	0.011 (1.21)
FirmOfcHides	0.000* (1.80)	0.006 (0.77)	0.019*** (3.96)	0.004 (1.00)	0.019** (2.20)	-0.006 (-0.72)	0.010 (0.33)	0.002 (0.25)	0.010 (1.44)	0.011 (1.17)
FundMgrHides	0.000** (2.00)	0.007 (0.79)	0.019*** (3.92)	0.004 (1.00)	0.019*** (2.69)	-0.006 (-1.20)	0.010** (1.54)	0.002 (0.29)	0.010** (1.58)	0.011* (1.70)
FullyHidden	0.000*** (2.87)	0.006 (0.49)	0.020*** (4.27)	0.004 (1.00)	0.020*** (1.67)	-0.005 (-0.44)	0.010*** (2.49)	0.002 (0.08)	0.010*** (2.59)	0.011 (0.81)

6.9 Table 9: Interaction intensity strengthens the alpha

Table 9 shows that when the manager and officer interacted on Facebook in the prior year, connected-stock returns are higher, especially for hidden ties. For FullyHidden, the interacted vs non-interacted long-short alpha is about 1.06% per month. Interpretation: evidence for an **active information channel** rather than passive familiarity.

6.10 Table 10: Abnormal returns concentrate on corporate news days

Table 10 moves to daily frequency and splits days into “news” vs “no news”. For FullyHidden, CH earns substantially higher daily alpha on news days than on no-news days, and CH minus NCH spreads are large on news days. Interpretation: the hidden tie premium is event-linked rather than a smooth, unconditional drift.

6.11 Table 11: Which news and which officers?

Table 11 deepens the mechanism:

- Panel A: the strongest spreads occur around **earnings/revenues** and **M&A** events.
- Panel B: performance aligns with expertise (e.g., CFO connections are particularly strong around earnings; M&A spreads also vary with executive role).
- Panel C: effects are stronger when analyst coverage is low, consistent with private information being more valuable when public information is scarce.

Table 11
Returns by news type, expertise, and analyst coverage.

	FullyHidden	Non-Connected	Long-Short	
Panel A: News Event Type				
Analyst Ratings	0.046 (0.53)	-0.009 (-0.13)	0.055 (0.56)	
Earnings and Revenues	0.234** (2.27)	0.030 (0.21)	0.203** (2.26)	
Capital Structure Changes	0.022 (0.21)	0.029 (1.21)	-0.007 (-0.08)	
Marketing and Investor Relations	0.076 (1.24)	0.008 (0.40)	0.068 (1.22)	
Mergers & Acquisitions	0.574*** (4.80)	0.184*** (6.98)	0.391*** (3.37)	
Other News	0.050 (0.85)	0.000 (0.07)	0.050 (0.87)	
Panel B: Firm Officer Expertise				
CEO	Earnings and Revenues	0.371 (1.63)	0.015 (0.13)	0.356 (1.62)
	Mergers & Acquisitions	0.422* (1.94)	0.103 (1.16)	0.319* (1.74)
	Earnings and Revenues	0.683** (2.31)	0.007 (0.09)	0.676** (2.27)
CFO	Earnings and Revenues	1.083*** (2.59)	0.208 (1.38)	0.875** (2.00)
	Mergers & Acquisitions	0.068 (0.59)	0.010 (0.18)	0.058 (0.56)
	Earnings and Revenues	0.108* (1.74)	0.099 (1.53)	0.008 (0.20)
Other C-suite	Mergers & Acquisitions			
Panel C: Analyst Coverage				
High Coverage	0.883* (1.90)	-0.037 (-0.45)	0.921** (1.98)	
Low Coverage	1.410*** (3.51)	-0.023 (-0.43)	1.433*** (3.57)	

Table 12
Cross-sectional Fama-Macbeth regressions.

	FullyVisible	FirmOfcHides	FundMgrHides	FullyHidden
Constant	0.0104*** [0.0038]	0.0108*** [0.0038]	0.0124*** [0.0038]	0.0111*** [0.0039]
$\beta(\text{DiffWeight})$	0.0142 [0.0088]	0.0152* [0.0089]	0.0171** [0.0085]	0.0228*** [0.0083]
ME	-0.0006** [0.0003]	-0.0006** [0.0003]	-0.0008** [0.0003]	-0.0007** [0.0003]
BM	0.0005 [0.0003]	0.0005 [0.0003]	0.0005* [0.0003]	0.0004 [0.0003]
MOM	0.0002 [0.0023]	0.0003 [0.0023]	0.0002 [0.0023]	0.0002 [0.0024]
STR	-0.0238*** [0.0057]	-0.0240*** [0.0057]	-0.0230*** [0.0057]	-0.0238*** [0.0058]
IMOM	0.0846*** [0.0187]	0.0830*** [0.0186]	0.0826*** [0.0190]	0.0839*** [0.0190]
SUE	-0.0009 [0.0012]	-0.0010 [0.0013]	-0.0006 [0.0012]	-0.0011 [0.0013]
Adj. R Squared	0.0680	0.0702	0.0783	0.0814
N	1,234,981	1,234,612	1,235,862	1,233,925
N Months	290	290	289	288

Table 13

Characteristics by Facebook identification and friends list visibility.

	Fund A					Fund B				
	Fb-identified		Non-identified		Diff.	Hidden Friends		Public Friends		Diff.
	Mean	SD	Mean	SD		Mean	SD	Mean	SD	
Age	44.43	3.53	47.67	2.29	-3.25***	43.52	3.91	44.98	3.41	-1.45*
Tenure	4.59	1.24	5.37	1.11	-0.79***	4.32	1.27	4.77	1.25	-0.46
Male	0.86	0.02	0.93	0.02	-0.08***	0.86	0.05	0.87	0.02	-0.01
CFA	0.56	0.06	0.54	0.07	0.02	0.58	0.08	0.56	0.07	0.02
MBA	0.61	0.05	0.59	0.04	0.02*	0.60	0.06	0.61	0.04	-0.01
Ivy League	0.33	0.09	0.30	0.09	0.02	0.33	0.11	0.32	0.09	0.01
Growth Fund	0.41	0.07	0.38	0.08	0.03	0.39	0.06	0.41	0.08	-0.02
Value Fund	0.20	0.03	0.22	0.07	-0.02	0.21	0.05	0.20	0.05	0.01
Small-Cap Fund	0.21	0.04	0.19	0.07	0.02	0.22	0.14	0.20	0.04	0.02
Large-Cap Fund	0.50	0.03	0.52	0.08	-0.02	0.51	0.08	0.49	0.04	0.01
Fund TNA (bn)	1.29	1.01	0.98	0.60	0.31	1.45	1.13	1.18	0.95	0.27
Age	47.50	4.86	53.68	2.18	-6.18***	46.01	5.62	48.39	4.77	-2.38*
Tenure	5.42	0.65	6.56	0.66	-1.14***	4.94	0.70	5.72	0.69	-0.78***
Male	0.80	0.07	0.85	0.04	-0.05***	0.79	0.09	0.81	0.06	-0.04**
MBA	0.28	0.04	0.27	0.01	0.01	0.26	0.04	0.28	0.05	-0.02
Ivy League	0.23	0.07	0.21	0.06	0.02	0.23	0.10	0.23	0.07	0.00
Firm Size (bn)	47.03	26.43	39.50	30.02	7.53	50.23	30.50	43.87	23.95	6.36

6.12 Table 12: Cross-sectional predictability—hidden ties are a powerful return predictor

Table 12 runs Fama–MacBeth regressions in Eq. (3). The key coefficient is on $\beta(DiffWeight)$, which is positive and becomes larger/more significant as ties become more hidden, peaking for FullyHidden. Interpretation: the hidden-tie signal predicts returns even after standard characteristics (size, value, momentum, reversals, industry momentum, SUE).

6.13 Table 13: Selection and observability checks

Table 13 compares (i) Facebook-identified vs non-identified individuals and (ii) hidden-friends-list vs public-friends-list individuals. The main role is to show that observable characteristics differ (e.g., identified individuals are younger on average), but these differences do not mechanically explain the core hidden-alpha patterns.

6.14 Table 14: Pre vs post Facebook inception

Table 14 shows the hidden-tie alpha exists in the pre-Facebook period (1984–2003) and becomes even larger post-2004. Interpretation: the effect is not “using Facebook to trade”; rather Facebook reveals pre-existing offline relationships, and the information value persists.

Table 14
Returns before and after Facebook's inception.

	Pre-Facebook four-factor alpha			Post-Facebook four-factor alpha		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	0.13 (0.26)	0.04 (0.08)	0.08 (0.21)	0.09 (0.47)	-0.01 (-0.18)	0.10 (0.49)
FirmOfcHides	0.27 (0.58)	0.01 (0.35)	0.26 (0.67)	0.34 (0.82)	-0.01 (-0.39)	0.36 (0.84)
FundMgrHides	0.49* (1.73)	0.02 (0.18)	0.47* (1.70)	0.79** (1.99)	-0.01 (-0.20)	0.81** (2.03)
FullyHidden	1.21** (2.15)	0.01 (0.06)	1.19** (2.12)	1.57*** (3.76)	-0.01 (-0.12)	1.58*** (3.79)

Table 15

Returns on connected stocks of index funds.

	Raw Return			Four-Factor Alpha		
	Connected	Non-Conn.	LS	Connected	Non-Conn.	LS
FullyVisible	0.87** (2.38)	0.88*** (3.33)	-0.01 (-0.09)	-0.03 (-0.25)	0.03 (0.85)	-0.06 (-0.12)
FirmOfcHides	0.84** (2.34)	0.82*** (2.97)	0.02 (0.12)	0.03 (0.31)	-0.00 (-0.06)	0.03 (0.28)
FundMgrHides	0.84** (2.36)	0.84*** (2.91)	0.00 (0.08)	0.01 (0.20)	0.02 (0.66)	-0.01 (-0.09)
FullyHidden	0.88** (2.41)	0.86*** (2.79)	0.02 (0.13)	0.02 (0.19)	0.02 (0.52)	0.00 (0.11)

6.15 Table 15: Index funds show no hidden alpha

Table 15 repeats connected-vs-non-connected tests for index funds. Long-short alphas are statistically insignificant across visibility levels. Interpretation: strategic selection and timing are necessary for the premium, consistent with an information-based trading channel.

7 Putting everything together

A narrative consistent with the evidence is:

- (1) Hidden personal ties provide a channel for **non-public information** to flow from officers to managers.
- (2) Managers **allocate more capital** to firms with such ties, increasingly so when ties are concealed (Table 5).
- (3) Only concealed ties generate strong outperformance (Table 6; Fig. 4).
- (4) Outperformance reflects **timing** (Tables 7–9; Table 8 is a key falsification against pure matching).
- (5) Returns are concentrated around information-intensive corporate events and in settings with less public information (Tables 10–11).
- (6) The signal is economically relevant cross-sectionally and is not a byproduct of Facebook identification or indexing constraints (Table 12; Fig. 5; Table 15).

8 Conclusion

8.1 Summary: what the paper concludes

This paper concludes that **concealed social ties** between mutual fund managers and public-firm officers are strongly associated with **economically large and statistically significant abnormal returns**, in a way that is consistent with **information-based trading through hidden networks**.

The paper’s core findings can be summarized as follows:

- (i) **Hidden ties generate large abnormal returns; visible ties do not.** Across the universe of mutual funds and public firms, stocks connected to managers through *hidden* manager–officer ties earn substantial abnormal performance. The headline magnitude reported in the paper is an average of about **135 bps per month** (exceeding **16% per year** in alpha; $t \approx 3.5$). In contrast, *publicly visible* ties have little predictive power for future returns.
- (ii) **The endogenous decision to hide a relationship is itself highly informative.** The strongest predictor is not simply “being connected,” but *how concealed the connection is*. The paper emphasizes that the **behavioral choice to conceal a relationship** dominates other observable network features in predicting future stock returns.
- (iii) **The premium is not explained by familiarity or selection/matching.** The paper provides evidence that the hidden-tie premium is not just “good managers happen to be connected to good firms” or “managers buy stocks of friends’ firms out of familiarity.” Instead, the abnormal performance is tied to managers’ **active decisions** about *when* to hold (or avoid) stocks linked to their hidden relationships.

- (iv) **Managers exhibit timing ability on hidden connections.** Managers earn higher returns on the connected stocks they *choose to hold* than on connected stocks they *could have held* but avoided, which is a key falsification against pure endogenous matching explanations. This indicates **timing/selection conditional on the connection**, not just unconditional connected-stock outperformance.
- (v) **The hidden-tie alpha concentrates around information-intensive corporate events.** The return premium is particularly strong around **earnings-related** news and **M&A** events, and it is stronger when the informational value of officer knowledge is plausibly higher.
- (vi) **The premium is absent in index funds (where strategic selection/timing is infeasible).** The absence of the effect in index funds supports the interpretation that the premium requires discretionary trading (strategic stock selection and timing) rather than mechanical exposure.
- (vii) **The hidden-tie signal remains robust across settings and persists through time.** The paper reports that the effect is robust across industries, firm types, time periods, and fund styles, and its predictive content increases over time through the most recent sample years.

8.2 Why does concealment predict “hidden alpha”?

The paper’s central intuition is that **concealment is an endogenous signal of information value and compliance risk**.

1) Information flow requires a channel; hidden ties identify that channel. A personal relationship between a fund manager and a firm officer creates a plausible channel through which *non-public, decision-relevant information* can be transmitted (directly or indirectly), especially about outcomes that move prices (earnings, acquisitions, major operational developments).

2) If the information is valuable (and potentially sensitive), parties have incentives to conceal the relationship. If a tie is associated with potentially privileged information flow, disclosure can raise scrutiny from compliance departments, investors, journalists, or regulators. Therefore, **hiding the relationship is not random**: it can reflect an incentive to reduce detection probability precisely when the relationship is most informationally potent.

3) The empirical signature of information is timing rather than unconditional drift. Pure “familiarity” would predict persistent overweighting without strong performance concentration. Instead, the paper finds that managers appear to know *when* to hold and when to avoid connected stocks, and the abnormal returns concentrate around news events. This pattern is consistent with a **timing-and-information** story:

hidden tie $\Rightarrow$ private signal about future cash-flow news $\Rightarrow$ select/timing trades
 $\Rightarrow$ event-concentrated abnormal returns.

4) **Why the effect disappears in index funds.** Index funds cannot materially deviate from benchmark composition, so even if the manager has information, they cannot convert it into abnormal returns. The absence of the premium in index funds supports the idea that **discretionary trading is required** to translate hidden-network information into alpha.

8.3 Takeaway

*The paper shows that mutual fund managers earn large abnormal returns primarily on stocks linked through **concealed** personal ties to firm officers; concealment is a powerful endogenous signal of information flow, and the resulting “hidden alpha” reflects active timing around information-intensive events rather than mechanical selection, familiarity, or simple matching.*